# MA1521 Calculus for Computing

## Lecture Group 2

## Lecturer: Prof To Wing Keung

## Lesson 18: Friday 16 October 2026

## (Week 9)

**Remark.** Q17.3 to Q17.4 are leftover exercises from the Tuesday class.

**Remark.** Q18.1 to Q18.3 are on Section 14.8 of Thomas' Calculus.

**Q17.3.** Evaluate the integral $\iint_R (x^2 + 2xy)\, dA$ where $R$ is the region bounded by the lines $y = 2x + 3$, $y = 2x + 1$, $y = 5 - x$ and $y = 2 - x$.

[Suggestion: From the equations of the boundary lines of $R$, get a transformation $u = u(x, y)$, $v = v(x, y)$, so that the region $G$ in the $uv$-plane will be described by very simple inequalities involving $u$, $v$.]

**Q17.4.** Evaluate $\iint_\Omega (x^2 + y^2)\, dA$, where $\Omega$ is the region in the $xy$-plane bounded by the lines

$$x + y = 0, \quad x + y = 1, \quad 2x - y = 0, \quad 2x - y = 1.$$

[Suggestion: Make a change of variables.]

**Q18.1.** Evaluate $\iint_\Omega xy\,dx\,dy$, where $\Omega$ is the region in the first quadrant bounded by the curves

$$x^2+y^2=4,\quad x^2+y^2=9,\quad x^2-y^2=1,\quad x^2-y^2=4.$$

[Suggestion: Make a change of variables.]

**Q18.2.** Recall that to compute double integrals using polar coordinates, we have

$$\iint_R f(x,y)\,dA=\iint_G f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta. \qquad (1)$$

Here $R$ is a region in the $xy$-plane, and $G$ is the corresponding region in the $r\theta$-plane. Use the change of variables formula for double integrals to derive (1).

**Q18.3.** Evaluating the following iterated integral by converting to polar coordinates and using the change of variables formula for double integrals:

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} xy \, dy \, dx.$$